Telehealth Hypertension Analytics

Explained, audited blood pressure staging from 24-hour monitoring

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1 Project Description

What it is. A decision-support tool that reads a patient's full 24-hour blood pressure record, suggests a hypertension stage, shows which measurements drove the result, and flags uncertain cases for a clinician.

Who it helps. Clinicians who review ambulatory monitoring (ABPM), where a wearable cuff takes about 48 readings across a day and night.

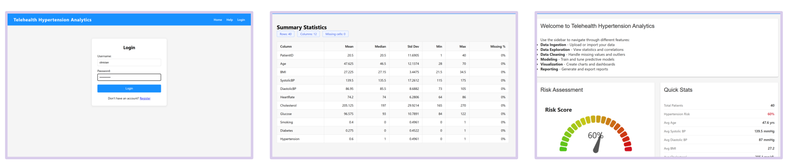
1 Record in
48 readings from one 24-hour cuff recording

2 Stage + explain
Normal, Elevated, Stage 1 or Stage 2, with reasons

3 Clinician decides
Low-confidence cases are flagged for review

What was built. A machine learning model, a secured web service (API), a dashboard page and 44 automated tests, released together as one tagged version on GitHub.

Data. Real patient records are protected, so every result here uses 1,800 simulated patients.



Screens from the running app: (1) role-based sign-in, (2) data upload with summary statistics, (3) dashboard with a risk score.

2 Project Justification

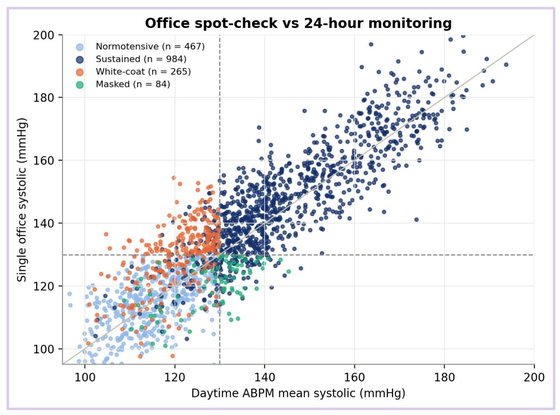
High blood pressure is a leading modifiable risk factor for cardiovascular disease, yet it is usually diagnosed from a few office readings (Whelton et al., 2018).

A single reading can mislead in both directions: white-coat hypertension looks high only in the clinic, and masked hypertension looks normal in the clinic but is high outside it.

25.5% one office reading disagrees with true status

54% office-rule staging accuracy

48 readings per patient to review by hand



Each dot is a simulated patient. Orange: high in clinic but not hypertensive (white-coat). Green: normal in clinic but hypertensive (masked).

3 Functional Requirements, Specifications and Standards

Requirement	Specification as built	Standard or evidence
Stage a patient from a 24-hour record	One API call takes about 48 readings and returns Normal, Elevated, Stage 1 or Stage 2	2017 ACC/AHA staging (Whelton et al., 2018)
Explain every result	Top four contributing measurements plus a plain-English summary	TreeSHAP attributions (Lundberg et al., 2020)
Give trustworthy confidence	Calibrated probabilities (error 0.014); results under 65% flagged for review	Reporting guided by TRIPOD+AI (Collins et al., 2024)
Respond within 200 ms	Measured 22 ms for prediction plus explanation	Project service-level objective
Protect patient data	Hashed passwords, role-based access tokens, rate limiting, encrypted patient columns	HIPAA Security Rule safeguards (U.S. Department of Health and Human Services, 2013)
Be reproducible and tested	Fixed random seeds, 44 automated tests, tagged release	Documented in a model card (Mitchell et al., 2019)

4 Design Process

Six stages, reshaped by two reviews: professor feedback on holdout leakage, then a self-audit.

1 Define requirements
Clinical need, staging rules and security needs set first

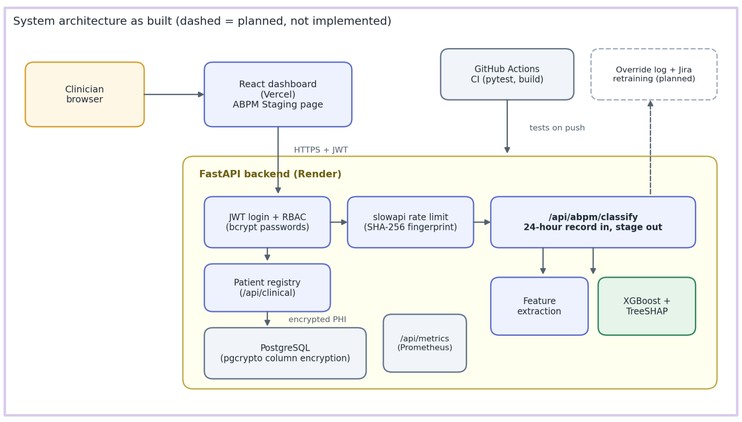
2 Simulate and prepare data
1,800 patients, 48 readings each, with noise, artifacts and gaps

3 Select the model fairly
Choices by cross-validation on training data only; 540 patients held back for one final test

4 Build the service
Secured API and dashboard page; model, model card and tests released together

5 Audit the whole project
Claims checked against the code; security and evaluation gaps found and fixed

6 Report honestly
Limits and open issues published with the results



System as built. The dashed box (clinician override logging and retraining) is planned, not implemented.

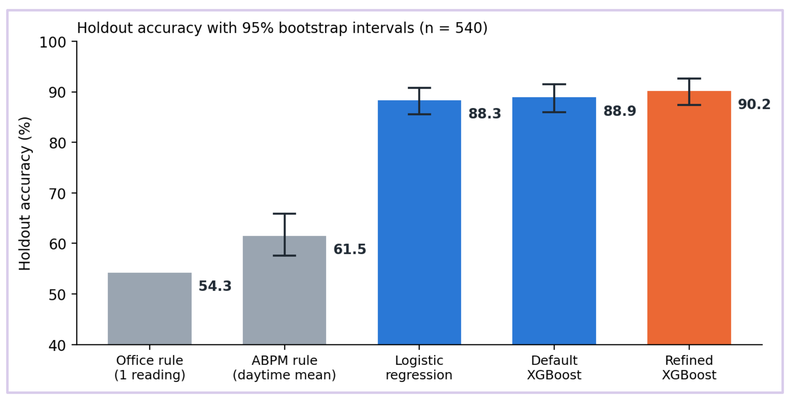
5 Relevant Data Summaries

90.2% of 540 unseen patients staged correctly

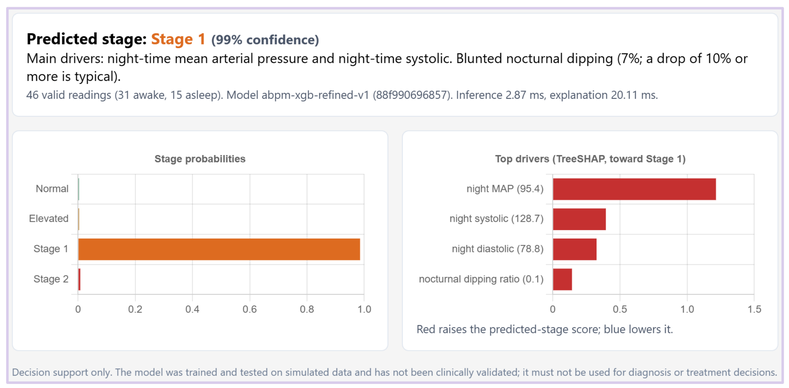
97% of Stage 2 patients caught (131 of 135)

0 patients staged more than one step from the true stage

22 ms to predict and explain one patient



Accuracy on patients the model never saw. Rules on one office reading or the daytime average are right 54% and 61% of the time. The gain over simpler machine learning is small and within the margin of error; the model's edge is better-calibrated confidence and explanations.



The dashboard's ABPM Staging page, captured from the running app: the predicted stage with its confidence, stage probabilities, and the four measurements that drove the result. Night-time pressure carries the most weight, a pattern built into the simulated data.

Where it errs. Most mistakes fall within 5 mmHg of a stage boundary, where the stage is a judgment call. Accuracy is lowest under age 45 (86%, 93 patients), so it needs a larger test.

6 Limits and Next Steps

- **Simulated data only.** Not clinically validated and not for patient care.
- **Next:** test on real multi-site records, predict real outcomes, and show prediction sets with coverage guarantees near stage boundaries (Angelopoulos & Bates, 2023).
- **Planned, not built:** clinician override logging and retraining.

References

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Code, notebook and tests: github.com/rajachakraborti/telehealth-hypertension-analytics | release tag topic7-submission (commit 73adefd) | rajachakraborti@gmail.com